ActInf Textbook Group ~ Cohort 7 ~ Session 17 (Chapter 6) ~ 2/27/2025

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all right welcome thank you everyone for joining it is the first early session for cohort 7 in this second phase of activity so anybody who uh wants to sort of just check in with uh that I guess we could have done before starting recording but just want to check in where are we at in this cohort as we head into the second half of this textbook in people's view or what would they like to do for this phase of activity for the kind of second half for this cohort and then we will turn to chapter six um maybe ali uh and then anyone else who wants to raise your hand or just go for it uh okay so uh chapter six actually marks the um uh beginning of the second part of the textbook uh as we probably already uh know the book organized into two different uh parts so part one is mainly theoretical and part one part two uh basically deals with more practical applications of active inference uh so as as we saw in the first part of the uh textbook um the theoretical Foundation of active inference is actually work in progress so uh it's not something uh that that has been uh uh completed uh in order to applied uh as a kind of uh comprehensive uh modeling tool but at the same time uh the second part of the uh textbook provides very promising road maps uh to um towards how we can apply even these uh tentative foundational work um in order to build uh some real World models of the intelligent agents or cognitive agents uh so I think that's important to keep that in mind because uh in many textbook groups uh I've seen people uh somehow uh rais this issue as to well it's not as comprehensive uh as we thought as as a modeling tool so uh since the public a of this book U obviously active influence research has progressed a lot uh but even the uh the theoretical foundations provided in the uh first part of the book uh I believe provides enough material to um at least build some uh Toy models uh that can capture some of the real world element of a real um a real world agents or at least cognitive agents thank you Ali great anyone else want to share with their ad as in this uh as we head into this part of the textbook yes so uh I think I would like to do that um I missed the first half of this um group uh so I'm I'm asking a question which belongs to chapter two I think or chapter three I don't know um and so my apologies for that and I don't want to hijack the discussion here uh and I've put my question also in the in the chat but basically comes down to um having tood the role of PR in um active inference as having a dual role which is one kind of the intuitive role for me at least being vely familiar with bence inference is you know encoding prior beliefs and also um having a kind of indicator role of encoding uh preferences and I think this comes out of uh discussion of um ocular motor Behavior as well as other things and I was just wondering if you have discussed this at all and uh if I've yeah if a brief comment on this would be great but if you say well you should have you know attended the previous sessions and I'm fully happy with that as well and then we'll just tag along for the right for now yes um there's a slide in this the point raised in this paper um that points to Prior being used as a um preference to drive pragmatic behavior in lie of for example what a reward function does for reinforcement learning because when things are framed in terms of likelihood directly rather than projecting onto a explicit rewarder utility uh function only just working with a base joint distribution um then the prior can be seen as preference constraint so it's like the preference for blood sugar has a different connotation and and um function and everything than one's felt psychological preference for something or revealed or stated preference but um at least um from a modeling perspective it's more similar than not okay so I'll I'll have a look at uh those reses thanks so okay Jerry yeah I think as as we're going into at least my area of interest and my questions are you know based in my field of work which is marketing and Communications and just how to apply this to you know um consumer Behavior you know to to actually build some models that use these principles and these Frameworks um so that we can have replicable studies um within this type of a field great well chapter six is a great place for that because uh uh it is a recipe set of questions with more in common than uh uh with just general uh statistical experiment planning and systems engineering and systems modeling broadly uh with a connection in specific ways to get to the kind of generative models that are described in basically chapter 4 and there's a lot of uh distance between where the this uh current text from 2022 will get you and the code running in your hands uh but that's a ground that is also rapidly being more traversible too and uh and so building out examples so that they're specific uh just you know there's there's a big gap there but that is it represents um in in a sense just how many degrees of freedom there really are in probabilistic modeling yep that makes sense and so yeah I'm excited to to dig into it it definitely seems like it's an area that uh is ready to be explored cool so I wanted to share this uh one little experiment this is from earlier today it's linked in the chapter page so I took the text of chapter 6 just plain text put it into this PL tax repo used cursor IDE to restructure it and obsidian link it so now here it is um with probably some light uh you know edits but has these links out to all these other topics also this repo you can just put a new file and say write a PDP type model for this because it has this whole knowledge scaffold that that we can also all contribute to so I think a as the uh years and the application Niche changes chap uh the textbook and chapter six like Ali said it will always be interesting to consider how it's like oh yeah just these matrices alone are kind of like a good representation of modeling this or that uh all the way on through however that gets friction and a lot of degrees of freedom in application when there are a few standardized open-source methods and it's unclear how to necessarily Implement just an equation into an actual program they like um or there can be multiple it just like oh why is there not just one KL Divergence function or like why don't we just have a variational free energy function there's sometime it sometimes it gets very uh data form specific computational environment specific mat lab Julia all these like things that come up in the application so I think that's one of the biggest changes between 2022 um as uh some will remember in terms of like we'll think okay before RX and fur pmdp mat lab now better augmented coding and knowledge engineering tools code generation tools um helping with some of that especially if we uh can collaborate I believe and have some people working with these code and program synthesis approaches to the structure learning and kind of uh sketching out more the Tex tree of the spaces of generating different aspects of specific uh generative modeling uh passages and then also people who are just like wanting to model or do inquiry into a specific like yeah do there have more squirrel crossing the bike path on Tuesday those kinds of uh test experiments so um but this is the application part it's almost like the textbook coming back to domain uh even though that kind of happens in chapter 5 to an extent so I hope that in this phase for we can uh possibly return to some of the code questions that have come up over the years like having code examples for um you know chapter 7 in chapter uh some of which we already have more uh Advanced things open source so it' be a subsetting not like a making but also these things just from being specified can be made very quickly to a first pass so if we want to to um develop the code and also the questions around going through application that would be really useful maybe maybe I could share my um project that I'm working on and then when it comes up relevant uh I'm Andre from Lithuania Andre kowas does anyone want to start with a specific question about chapter 6 or the book first and then then let's do projects but not um begin with it thank you address what should we really do in synchronous asynchronous for the textbook the 2022 textbook is a great art artifact understand and learn from I mean any section or any question can be useful um though really I am I just nice nice drawing hey Daniel um yeah at least you're just asking questions there like the the one up there about affordances is always of interest to me as well as you know maybe how like variational script might come into play if that's applicable as well yeah what what about it um I mean I think at least from from my perspective um there was some stuff I'd read Around creating fields of affordances which I felt kind of tied from a consumerism perspective to how Brands go out to market for instance um and so maybe the question is more around like not just affordance generally but that idea of like how do organisms in this model or how would we test um how agents select between the field of importances through those either I don't know if that's through policy selection there that's my assumption but um you know still newer to thinking through this from a like actual application perspective does anyone want to say something to okay the first question in the for four question recipe here is what's this system are we modeling and then if we were thinking about an individual uh Mouse in the tmas like what's going to come up in chapter 7 in discrete time it's like okay are we're going to model it with four locations starting left right and down uh or are we going to model its footsteps so if you model footsteps then there's going to be the the explicit you're going to have some data structure that represents the footsteps location like data about it Andor simulated distributions about it but if you're going to just say well not it's not a footstep modeling project it's just a a location for state that which is what's going to happen in chapter 7 so um just from this sort of Applied angle which system are Remodeling and that's kind of reflect that the earlier comment about this having more in common especially early in this process with systems modeling and systems engineering it's just like okay the organism is a very complex Locus person at a game machine or something but but um like the complexity of how hormones interact with uh you know sound and all this or something or microbiome those are all part of the real environment SL situation SL system but you might be able to get away with a 3X3 Matrix for just three uh states of vigilance low medium high because those kinds of modeling uh that's just oh all models uh map territory all those kinds of topics that that come back and but it's like right it's a it's a model of this it's an inferred parameter it's a risk factor for x or it's just a mutual information estimate on percentile on this in our model so it that it that's why this whole process where chapter 6 actually gets one isn't necessarily some um G paradigm shift on their modeling view per se that you couldn't necessarily also have from doing a regular gen generalized linear modeling which can also be very sophisticated and deal with higher order um variance and causal relations it's kind of like generalized probabilistic modeling combined with cognitive syst systems approaches to systems engineering okay yeah that makes sense in the footstep model when you brought that up I think that kind of resonates for me because we'll talk a lot about customer Journeys and one thing I've been thinking about is just you know using the T the term consumerism like I don't know that we think about it the right way in my industry and so I've been looking to biology a little bit for a better analogy and so like predatory behavior for instance like the process of identifying and detecting and consuming and then processing something I think is an interesting corollary to what we probably ought to be doing when we talk about you know communication and purchase and stuff like that so that's that's interesting and helpful um and maybe that's something I can dig a little bit more into is that idea of more modeling footsteps versus locations which I think we tend to try to model the end the location when we ought to be modeling like the steps towards that yeah so the the hierarchical models at least in principle which is kind of where the second half of the Tex goes is is um where you have slower inference models you know and then a faster ones and then uh it's just a project or situation specific empirical question about how much compute and what kinds of specific models you're going to make um so so this let's continue on as people if if they want to bring in their um you know how they're seeing that or some other section of the of six or we can go to the questions but just yeah this first question is like is like okay we're modeling everything about the Biometrics of the person or just this those are such different questions but those are questions that are also just dealt with everyday in animal behavior cognitive modeling okay but here's where we start getting more into some of the technical analytical details of the structure of these models that relate to making some more committed model decisions not not just agreeing that okay we're we're just interested in um this question about this system of interest um so that that the next three questions are the three practical challenges these can be specified um like they go into more detail in chapter 6 and there's there's more sort of links into guides and experiences people can share and and just try and explore uh through but basically it's like keeping in mind that the textbook here uses generative model to describe the agent type uh unit and the generative process to be the environment um whereas also generative model can be used just to be the total um model specification basically the three um areas of questions are how do you want to deal with time um model it discreetly like or in continuous time uh these are questions common to other time series modeling approaches um even if you choose continuous time you know how will time discretization on if Will that matter for your data or for your compute um what is going to be the structure um in hierarchy [Music] um laterality like just message passing is it just like are you modeling one abstract decision maker making one sequential response to one put data just as an intuition pump or is it doing like this data to here and wanting this output so that's just sort of the um getting to specifics about the model structure which can again just be either specified uh for from Simplicity like just be often the models are just a single uh PDP partially observable Markov decision process and that that um can't necessarily be directly picked up to work on a large data set or something like that but for certain purposes it might be all the research question needs at that stage but um again that's kind of a separating like okay we know the systems have hierarchical depth so what parts of multiscale complexity are modeled so that's just a common modeling question and then what is the time how do these different agent or generative model specifications themselves deal with time so just some it's uh things that come up but these are more in common with uh at this point like statistical modeling causal inference machine learning AI also these kinds of questions would come up here is where it starts to get more into um some of the uh basian uh degrees of freedom of modeling like are you going to model a parameter as just fixed like a a spike in the probability space like not learnable you know how learnable how variable and how what are you going to do with how you specify and update and fix how learnable and how variable are you just going to say from this previous study we know the variance of this trait to be 10 so we just fixed the variance at 10 or we learned it but um you know if everything is learnable there's just the Looms a greater problem just like in uh uh reward learning when when they get into meta learning and all this how learning how fast to learn to learn and all the sort of things is um I mean the larger the state space learnability of the the of the kind of Target Model you make an optimization challenge by making it so open-ended by dealing with large data by dealing with complex data maybe there are some structures of probabilistic models that have excellent computational performance on certain situations but that's a hope and a preference and uh is not the case that that that uh you know any given uh probabilistic model would be performant in any given way on any given data set however there might be some that are really really effective or or the huge families that are adequate for certain uh uh processes but but but we're we're getting into statistical modeling so that's kind of from coming coming to this from an empirical biology behavioral modeling perspective and seeing how mostly here we're just talking about how some this could be said and probably many parts are said across different cognitive and behavioral modeling works but it's working within a specific probabilistic modeling framework that has connections to the free energy principle um sort of path of least action approach coming from The High Road chapter six sort of brings us back to the low road and building up the specific systems that um also have the extension into the implementation Methods at least uh some as we sort of explored a little earlier but it um getting from the uh questions about the specific system to the probability distributions in families and mod model that that meet up in the Middle with the sort of high road approach is basically where chapter 6 is but there's no it's like these questions are all common to statistical modeling efforts and there's other things I'm sure people here too could add of just the sort of um her istics and pitfalls and important questions for interpretation of complex systems modeling another sort of I mean another uh practical question is like um where what is the what there's different ways to do projects and and research and everything but does someone want to just imagine a model where there's a 10 time Horizon and a five of this and it takes in this data it's like is it just being sketched as part of a thinking process or is this wanting to be used for doing statistical modeling because the sometimes there's moves that are possible in just sort of drawing connections on paper that might be very uh there's that might be like hard within current implementation approaches so in the examples of of the code we kind of are building out um motifs that are useful but the the way that people are thinking about systems um is often very nuanced and so it's I I I mean I I hope people can in the second uh phase here like let let uh figure out how to make it useful also just discuss what do we really want to do with these meetings with recorded and unrecorded part because we we we can figure out um other ways to have our shared attention and work I don't want to uh feel like returning to this and just re-recording going back to the same if that's not the most useful format this to kind of break the to kind of break the silence here I'm not PR sure because like what um what what do you usually do like what is your usual um mode of like navigating uh the second half of the book like the Practical part of the book um like for instance someone uh like I think Andreas was about to sorry I was about to talk about his project I think hearing about uh like the projects that people are working on like how they're implement all this stuff that would definitely be sorry that would definitely be interesting for me um also for my own project like I don't have anything specific yet but for the coming meetings I think it would definitely be interesting to um to discuss the specifics of that but I'm I'm not sure what other things you you would yourself suggest J Daniel or like what has worked for uh previous previous meetings but that would be interesting for me thanks you yes okay we can return that's a good suggestion for Andreas um in terms of what has been useful people can look through the previous chapter 6 uh meetings but I I think what would be very useful is to uh connect specific questions and ideas that people have to segments of the text if only as an exercise in anchoring and and Scaffolding off of the 2022 text um maybe just as a jumping off point in moving to other resources then uh working through these questions and figuring out what other questions they asked along the way to get to they wanted to go with a probabilistic model that that they um could share uh back and uh connect to implementations uh tables that that we have to the textbook examples so sort of it to if we want to make models in given domains let's make them over these coming months according to this recipe Andor making our own but uh at least taking this recipe uh seriously as or even as a subset of things that we do Andreas how about just go for your question let's see where it goes yeah uh so um um to introduce myself I'm Andrew kowas from Lithuania I learned about active inference a year ago from Daniel and I've had many conversations with him I've read the first six chapters and so I'm attending at this time U every week uh and so what I'm interested in modeling um I'll say generally and then specifically generally I have this uh I've been documenting my own human expence uh as a conceptual language and so I'm uh one way that comes out is like I I feel like I live my life through three Minds there's an answering mind a questioning mind an investigating mind so like if I raise my hand there's me that knows what to do and I just tell myself to raise it and I don't have to think very much about how it's being raised or where it's being raised so and similarly like there's all kinds of answers that come into my head so that's the unconscious um these answers and then there I have another part of mind that's this chattering language uh words or or concepts but I think of them as questions so like when I think of like the word horse it's really a placeholder for what the answering mind might tell me as a horse and so I have this conceptual language and then I have um this full-fledged Consciousness which I call the investigating mind willful cognizant deliberate which basically tries to um line up the other two because get the same information in two different forms so the unconscious knows all these things the unconscious doesn't know them and so can have the information that two and then like with conom okay I apologize can you hear me yeah continue I apologize on so I was talking about this investigatory mind which aligns the other two what we know what we don't know and then like with Daniel conman's Thinking Fast thinking slow it's able to decide hey here I should just think fast and and where here is here I should think rationally you know so go with my Intuition or should go rationally I'm trying to model that so specifically how could I model especially and I think there's a language for each of those like so there's a language of how things come to matter for the unconscious there's a language of how meaning arises like verbalization for the conscious and there's a language for how okay not sure if his connection staple um yeah Danel to the last part he was talking about I I think I have a question maybe tack on to that about like when he's talking about system one system too yeah does that in some way tie to like how we would try to model novelty versus habit um in terms of like you that's something I've been thinking about as well like a lot of behavior that we notice in my field is is habit but we're trying to catch attention I think through novelty um to train something new in there yeah like um to that um and also a little bit about between habit and deliberation which Andre I hope we'll start to get at some of these points that that we can continue to um work on and develop in here in chapter 5 so not to go back but um chapter 5 reviews the some of the key model motifs from the Maman neuroanatomy cognitive modeling literature um studying people's brains H that being where friston and others have focused a lot of their work um so as sort of like well this is sort of what it looks like to study a complex system uh like the brain on this so here are two different um extrema modes of how policy priors like how apt one is to do something before receiving the next observation and uh policy inference uh getting to a policy posterior which action is sampled from so it's like the differencing of what was apt to happen before to what was apt to happen after for action and these are the distributions that are calculated during policy uh inference planning as inference actually implementing planning agents you will have one um or the other or a modulated blend of these as as it turns out to be for the brain for policy inference on the left is a habit passed through mode which is like cached action preferences which can be conditionally set on the right is the expected free energy calculation which is uh discussed more in the first half of the book and we can look at implementations in PDP or in uh other implementations in this phase of activity but this is where specific counterfactual policies of a given temporal Horizon are rolled out like policy research and evaluated in terms of expected free energy or free energy the expected future um so explicit policy roll out reweighted energy based learning consideration of epistemic plus pragmatic value of policies um know large computational blowup just like any other tree based uh policy search so that's where um secondary teristics like um bounded TR searches and policy switching and things like that uh come into play for for making these models large but often again it's only a simple one with four policies being considered or something or 16 policies being considered but but this has that sort of multi combinatoric um computational complexity if explicit probabilistic or to the extent explicit probabilistic rollouts are used um and then on the left is like basically just a pass through so broadly that connects to the thinking fast and slow and I think I think some papers have made some um points but it just sort of a it's a functional uh stratification in cognitive systems that and then that is being here modeled by one switching parameter not map and the territory all that it's like could there be two switching parameters but that's the whole experimental biology question and I would jump in to say like with marketing um there's very much the opportunity like you can sell to somebody's reflexes or you can sell to somebody's rationality and if you're a buyer you know you kind of want to have both satisfied you want to feel good um instinctively and you want to feel good in terms of you know your rational judgment uh and so I apologize for my bad connection but I wanted to say um what I want to model specifically I think would be um uh this language of how things come to have meaning which is relevant for the rational mind so because the mind that just works instinctively it just doesn't need to know all these things but once you introduce this second rational mind it needs to think both you know has to kind of be aware of both ways and kind of do do its best with this other mind so I thought well mathematics is something where things don't really matter but the meaning arises all the time and I thought I would look in particular to a very narrow field which would be triangle geometry and it turns out in that field there's an encyclopedia of triangle centers and basically like everything interesting about triangles is basically like a center of a circle so you can build that center of a triangle so to speak in 60,000 different ways but to to focus on the very simplest way um if you put a circle in a triangle it doesn't matter what the shape of the triangle is but um if you expand it to the maximum size there will be three points that it intersects that it touches on the triangle and the property will be because the radius of the circle is fixed that from the center of the circle it will be equally positioned you know distance-wise to each of those three touching points so that's called the in circle so one way is to try to imagine these two minds as communicating that problem so the unconscious mind kind of imagine it it places the circle in the Triangle you know you're giving a random triangle and you're kind of like blind so the unconscious mind places a circle in the triangle and it's allowed to expand the tri the circle until it touches something and it has to stop and mathematically it'll only touch one side basically like probabilistically it's impossible to touch more than one side at a time and then the conscious mind will say okay I want you to move but let's say it can't it can give an instruction to to move but uh the motion will be random in some random Direction so or I guess it has to move it has to move away from the it has it only has one direction it has if it doesn't want it has to move away from the line so it just moves away from the line at a certain speed let's say that's based on the rad proportional to the radius and then it will the conscious mind tells it how far to move like how many radi to move or more like percentage of radi and then it can expand again it knows how to expand the unconscious mind and it'll bump into something else maybe the same side maybe another side and so through this iterative process at a certain point it will have less and less room to move it'll never know what side it's bumping into because it doesn't have that information but it'll just know that it's starting to bump in very often on you know which would presume that it's being closed in and so then the idea is that once you get within whatever accuracy you want because it never touches all three sides at once it only touches one side at a time um realistically but um when you get within 1% or whatever you want then you say okay I'm happy I'll stop and so then you can ask for example well what can you deduce about the shape of the triangle based on the interactions that you had blind like this you can also have like a learning process you know then you can maybe compare that with the original triangle uh you can um maybe also have a um a learning process where the conscious part that decides um how far to go like how how far to move this um it can um different policies can yield different results so it can maybe try to learn that so I don't know if that was like a my initial attempt to try to come up with a way of thinking about this language and seeing like how active inference could kind of help to apply it yeah at least I think that's interesting what you were saying and it there's some stuff I was reading by Michael Levan around like an axis of persu with different nested levels of intelligence that's kind of interesting to me that I think touches on what you're you're saying about different you know I guess lower levels of um the system learning something and sending a signal out globally when there's some sort of an error um but I've been thinking about that those levels of persuasion because I think it speaks to from a communication perspective the way that we communicate we do it through unconscious associations we do it more rationally we do it emotively from a cultural perspective um but it like you know I think when it's most effective it's across all of those levels or that axis of persuasion that he's kind of talking about so is there such a thing as one of these models that's so simple it could be coded from a blank sheet of paper in I guess five minutes yes that's what the textbook level focuses on the textbook version is just five matrices with it in chapter 7 that's what this is what you build up to it's just starting from um Dynamic perception models partially observable models observations coming in this could be a sing binary variable this could be a three-state variable this could be continuous V like that's the work of modeling but then you have a hidden or latent State temperature in the room temperature on the thermometer mapping Matrix initial prior to kick the chain off transition Matrix then that's that's equivalent to a Colman filter like modeling then you add in some element of policy selection is it just a binary uh with two fixed outcomes or do you want to model this or that uh conditionality or learning or or or uh affordance State St space changing but you know those are sort of those are those are structures of models families of models that you may on your on your Intuition or or uh fun or whatever seek to just work within a given model family just because or not realize um that you're with within a given constraint um but it's just what is it is it just being analytically just here's like the space of all possible policies of the organism and I'm about to make a philosophical point or we chose to just model the blood sugar not engaging with other biomarkers but this is these are the matrices and the variables that you get to a b c d well I'd sort of like to see this implemented in code uh is that available yeah great question that's um again that's what chapter 6 gets to and I'll just link I I know it's in the textbook group um somewhere hold this is a table of of implementations if people want to add and curate more it would be awesome there's always uh a lot we can add with these repos um and if you again these are for people who want to contribute open source resources Andor review them and annotate them with um like what features they have like which ones have planning which ones use this or that uh which ones have this C I just want like a a really really simple one you can figure out what it's doing by inspection yeah isn't doesn't appendix C mention like the T thing that you were talking about the t- maze is there is there code for that um there is mat lab code for that it's a uh in appendix C but we haven't in a prior textbook group ported it but if people would like to uh work on that in in the times between um meetings and uh get it working or just share what they where they got to that could be awesome but making PDP models and using expected free energy and variational free um in the implementations and in this subsidi um repo in the cognitive repo there are there are it's just pdps with vfe and EF also I want this is the slide from the um the the three PS a distinct rule for the distribution P this is the same underlying prior distribution so if we were modeling like we're move we're interested in the movement data of thermophilic bacteria so we're modeling the distribution of their location with respect to some fixed wall so we're going to have some distribution for their location it is in the first um frame the probability of they're existing at that location at that measured time it's a it's a latent distribution smooth analytically um approximable Etc variational distribution implementable with software tools for their location it is also from the co cognitive behavioral bacterial side as their preference like our um expected pragmatic homeostatic firstness and tendency to um of certain variables to return to a TR States and it's the Fitness in that it's the uh ecological sampling availability constituting differential persistence phenotypically or genotypically but that just just that's one it's like we're doing a Pon distribution from the wall any closing uh comments if if people want to think about what will make this early time slot in its alternating you know next week at this earlier time slot we will go to chapter 2 so if people are um want to just stay at the same time continuing in this way we've done it or just do people have certain amounts of time they want to set aside or certain Milestones they want to reach then uh I I mean I want it to be useful we can make generative guides and material absurdly with people's feedback and help if they want to make certain kinds of models so it be it would be awesome to work on that um but if people have some uh preference for for what would be good given that we all are here uh for this time let's figure out so just like email uh or or at the next meeting add a note or we'll discuss it oh I just like to say I think uh some more like coding Workshop type elements would be good yeah I I I totally agree I'll be trying to code what I'm doing so and I'm interested to talk with others about you know how they code and yeah it's like the low road is like the show me the code Road high road is like some is um symbolic and analytical and axiomatic uh then sort of qualitative philosophical stances are included in that sort of combination and uh I think we can we can have some cool code models please just surface whichever uh desada you see and let's uh do it so I wanted to thank uh for the time I apologize for my connection and just to say like what I learned was that like what am I trying to predict I'm predicting where I think the center of the triangle is and so when I move the circle from The Edge I have to make a prediction how far to move it and that prediction could be wrong you know and I get feedback from the from the from the perimeter of the circle so it's a conversation between the predictions for the center of the circle and then the feedback that you get from the boundary of the circle that's getting closer to I think so and I guess for active inference you really need to understand like prediction eror like what is the prediction what's the source of the error just random point about this circle triangle thing wouldn't you want to after it hits the wall wouldn't you want to grow it in a particular direction rather than move it in a particular direction well I want to have that dialogue so um so um the the two steps are you can move you can grow so the unconscious mind only knows how to execute this it doesn't and it knows when it hits something but it doesn't know anything really so this blind conscious mind has to has to have this uh communication that's what I want to model might it be better to have grow from the center and grow in a particular direction when well the point is that you don't know where the center is you're trying to find the center you're randomly inserted wait I thought you knew the center of the circle just not the triangle oh the center of the circle you know right so and you can only expand you can grow larger but you can also grow a circle in a single Direction rather than from the center of the circle and and I think that would be a faster implementation of what you're yeah I'm not trying to do it faster

I'm trying to do slower so stupider like what's the stupidest implementation that's what I'm trying to do most stupid in revealing thanks though okay thank you bye